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United States Patent	12409262
Kind Code	B2
Date of Patent	September 09, 2025
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Plasmapheresis centrifuge bowl

Abstract

A plasmapheresis bowl for the separation and collection of plasma includes a core and a feed tube that increase the bowl efficiency and reduce foaming within the plasma. The core may have a cylindrical body and a ledge located within the interior of the core. The ledge extends radially inward from the core and defines, at least partially, a collection chamber within the plasmapheresis bowl. The core also has ribs that extend above the top of core body and create flow paths that allow fluid to enter the interior of the cylindrical body and collection chamber. The feed tube has a flow path extending through it that fluidly connects an inlet port on the plasmapheresis centrifuge. A first skirt member on the feed tube has a smooth angled surface that helps to reduce foaming.

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Appl. No.:	17/349966
Filed:	June 17, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20210308353 A1	Oct. 07, 2021

Related U.S. Application Data

continuation parent-doc US 15935476 20180326 US 11065376 child-doc US 17349966

Publication Classification

Int. Cl.: A61M1/36 (20060101); A61M1/34 (20060101); B04B5/04 (20060101); B04B7/08 (20060101)

U.S. Cl.:

CPC A61M1/3696 (20140204); A61M1/3496 (20130101); B04B5/0442 (20130101); A61M1/3403 (20140204); A61M2202/0415 (20130101); A61M2205/3306 (20130101); B04B2005/0464 (20130101); B04B7/08 (20130101)

Field of Classification Search

CPC: A61M (1/3496); A61M (1/3696); A61M (2202/0415); B04B (2005/0464); B04B (5/0442); B04B (7/08)

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Background/Summary

PRIORITY (1) This application is a continuation of and claims priority from co-pending U.S. application Ser. No. 15/935,476, entitled “Plasmapheresis Centrifuge Bowl,” filed Mar. 26, 2018, and naming Taylor Korytko as inventor, the disclosure of which is incorporated herein, in its entirety, by reference.

TECHNICAL FIELD

(1) The present invention relates to systems and methods for blood apheresis, and more particularly centrifuge bowls for collecting a plasma product.

BACKGROUND ART

(2) Apheresis is a procedure in which individual blood components can be separated and collected from whole blood temporarily withdrawn from a subject. Typically, whole blood is withdrawn through a needle inserted into a vein of the subjects arm and into a cell separator, such as a centrifugal bowl. Once the whole blood is separated into its various components, one or more of the components (e.g., plasma) can be removed from the centrifugal bowl. The remaining components can be returned to the subject along with optional compensation fluid to make up for the volume of the removed component. The process of drawing and returning continues until the quantity of the desired component has been collected, at which point the process is stopped. A central feature of apheresis systems is that the processed but unwanted components are returned to the donor. Separated blood components may include, for example, a high density component such as red blood cells, an intermediate density component such as platelets or white blood cells, and a lower density component such as plasma.

(3) Some of the currently available centrifuge bowls are subject to turbulence and shear forces that negatively impact blood component separation and plasma collection. For instance, some prior art centrifuge bowls allow turbulence and shear forces (e.g., generated by spinning fluid contacting an effluent skirt) to be transmitted into the separation chamber of the bowl. This, in turn, disturbs the separation of the cells in the separation chamber, causes a noisy bowl optics signal, and reduces cellular separation from the plasma. Additionally, the turbulence and shear forces may cause foaming within the plasma that is to be collected.

SUMMARY OF THE INVENTION

(4) In accordance with some embodiments of the present invention, a core for a plasmapheresis bowl may include a cylindrical body, a ledge, and a plurality of rib members. The cylindrical body defines the core and an interior of the core. The ledge may be located within the interior of the core

between a proximal end and a distal end of the cylindrical body, and may extend radially inward from the inner diameter of the core. The ledge may have a top surface that defines, at least partially, a collection chamber within the plasmapheresis bowl. The ribs may be located on and spaced about a proximal portion of the cylindrical body nearer the proximal end. The ribs may extend above the proximal end of the cylindrical body and create flow paths between them that allow fluid to enter the interior of the cylindrical body and the collection chamber. The plurality of ribs may include three or more ribs (e.g., eight ribs).

(5) In some embodiments, the cylindrical body may have a second portion that is located below the ledge. The second portion may provide a reflective surface for an optical sensor. Additionally, the cylindrical body may have a distal portion located below the second portion that may stabilize fluid within the plasmapheresis bowl (e.g., during processing). The proximal portion may be located above the ledge, and may isolate the collection chamber from a separation region within the plasmapheresis bowl. Additionally or alternatively, the proximal portion may prevent turbulence and shear forces within the collection chamber from reaching the separation region. The inner wall of the proximal portion may define a side wall of the collection chamber.

(6) Each of the plurality of ribs may include a top surface, a bottom surface, and an outer surface. The top surface of at least one of the ribs may interface with a portion of the plasmapheresis bowl to secure the cylindrical body within the plasmapheresis bowl. The bottom surface of at least one of ribs may interface with a mating ledge on a body of the plasmapheresis bowl to locate the core in the plasmapheresis bowl. The outer surface of at least one of ribs may interface with an interior surface of the plasmapheresis bowl (e.g., within the neck portion of the bowl) and create an interference fit between the core and the plasmapheresis bowl.

(7) In some embodiments, the top surface of the ledge may slope downward toward the distal end of the cylindrical body. The ledge may also include a bottom surface that prevents fluid within the interior of the core and below the ledge from entering the collection chamber. The ledge may also include an opening that extends through the ledge. The opening may allow a feed tube of the plasmapheresis bowl to pass through the ledge and allow fluid within the collection chamber to exit the collection chamber when the plasmapheresis bowl is stopped. The opening may be located at the center of the ledge such that it is coaxial with the cylindrical body. The cylindrical body may have a constant outer diameter along a length of the cylindrical body.

(8) In accordance with additional embodiments, a feed tube for a plasmapheresis centrifuge bowl may include a tubular member extending between a proximal portion end and a distal end, and a first skirt member. The tubular member may have a flow path extending through it that fluidly connects an inlet port on the plasmapheresis centrifuge bowl and an interior of the plasmapheresis centrifuge bowl. The first skirt member may extend radially outward from the tubular member and may have (1) a first surface that is generally perpendicular to the tubular member and (2) an angled surface extending radially outward and distally from the first surface. The angled surface may be smooth.

(9) The feed tube may also have at least one spacing rib (e.g., three ribs equally spaced about the skirt member) that is located on the first surface. The spacing rib may space the first skirt member from a second skirt member to create a fluid channel extending between the first skirt member and the second skirt member. The second skirt member may be located on a header assembly of the plasmapheresis centrifuge bowl. The spacing rib(s) may have a first portion that extends along the first surface, and a second portion that extends proximally along at least a portion of the tubular member. A core within the plasmapheresis bowl and the first and second skirts may be configured such that the distance between the inner diameter of a proximal portion of the core and the outer diameter of the first and second skirts is maximized.

(10) The fluid channel may fluidly connect a collection chamber within the plasmapheresis centrifuge bowl with an outlet port on the plasmapheresis centrifuge bowl. The feed tube may also include an extension tube that is connected to the tubular member at the distal end and extends

toward a bottom of the plasmapheresis centrifuge bowl. Fluid entering the plasmapheresis centrifuge bowl via the feed tube may be introduced nearer to the bottom of the plasmapheresis centrifuge bowl.

(11) In accordance with further embodiments, a plasmapheresis bowl may include an outer body that is rotatable about a longitudinal axis of the centrifuge bowl. The outer body may have a main body defining an interior cavity, a neck portion extending proximal to the main body, and a shoulder connecting the main body and the neck portion. The bowl may also include a core located within and rotatable with the outer body. The core may have (1) a cylindrical body defining the core and an interior of the core, (2) a ledge, and (3) a plurality of ribs (e.g., eight ribs). The ledge may be located within the interior of the core between a proximal end and a distal end of the cylindrical body, and may extend radially inward from an inner diameter of the core. The ledge may have a top surface that defines, at least partially, a collection chamber within the plasmapheresis bowl. The ribs may be located on and spaced about a proximal portion of the cylindrical body nearer the proximal end of the cylindrical body. The ribs may also extend above the proximal end of the cylindrical body to create flow paths between each of the ribs. The flow paths allow fluid to enter the interior of the cylindrical body and collection chamber.

(12) The bowl may also have a separation region located between the core and the outer body, and rotation of the centrifuge bowl may separate the whole blood within the separation region into a first blood component and a second blood component. Additionally, the bowl may have an inlet port for introducing whole blood into the plasmapheresis bowl, an outlet port for extracting a first blood component out of the centrifuge bowl, a feed tube and a rotary seal. The feed tube may be fluidly connected to and extend distally from the inlet port toward a bottom of the outer body and may introduce the whole blood into the plasmapheresis bowl. The rotary seal may be part of the header assembly which may be attached to the outer body and fluidly couple the inlet port and outlet port to the outer body.

(13) In some embodiments, the cylindrical body of the core may have a second portion located below the ledge and at least a portion of the second portion may provide a reflective surface for an optical sensor. Additionally or alternatively, the cylindrical body may have a distal portion that is located below the second portion and is configured to stabilize fluid within the plasmapheresis bowl. For example, the distal portion of the cylindrical body may be configured to stabilize a plasma layer within the separation region. The inner diameter of a separated plasma layer within the separation region may contact the distal portion of the cylindrical body.

(14) The proximal portion may be located above the ledge and may isolate the collection chamber from the separation region. For example the proximal portion may prevent turbulence and shear forces within the collection chamber from reaching the separation region. The inner wall of the proximal portion may define a side wall of the collection chamber.

(15) Each of the plurality of ribs may include a top surface, an outer surface, and a bottom surface. The top surface of at least one rib may interface with a portion of the seal crown to secure the cylindrical body within the plasmapheresis bowl. The outer body may have a mating ledge within the neck portion and the bottom surface of at least one of the ribs may interface with the mating ledge of the plasmapheresis bowl (e.g., to locate the core in the plasmapheresis bowl). The outer surface of at least one rib may interface with an interior surface of the neck portion of the plasmapheresis bowl to create an interference fit between the core and the plasmapheresis bowl.

(16) In further embodiments, the top surface of the ledge may slope downward toward the distal end of the cylindrical body. The bottom surface of the ledge may prevent fluid within the interior of the core and below the ledge from entering the collection chamber. Additionally or alternatively, the ledge may have an opening extending through it. In such embodiments, the feed tube may extend through the opening. The opening may be located at the center of the ledge such that it is coaxial with the cylindrical body. The cylindrical body may have a constant outer diameter along a length of the cylindrical body.

(17) The feed tube may include a tubular member and a first skirt. The tubular member may extend between a proximal end and a distal end of the feed tube, and may have a flow path extending therethrough. The flow path may fluidly connect the inlet port and the interior cavity of the plasmapheresis bowl. The first skirt member may extend radially outward from the tubular member. The first skirt may have first surface that is generally perpendicular to the tubular member and an angled surface extending radially outward and distally from the first surface. The angled surface may be smooth.

(18) In additional embodiments, the feed tube further may include at least one spacing rib (e.g., three ribs that are equally spaced about the skirt member) located on the first surface. The spacing rib(s) may space the first skirt member from a second skirt member to create a fluid channel extending between skirt members. The fluid channel may fluidly connect the collection chamber and the outlet port. The second skirt member may be located on a header assembly of the plasmapheresis centrifuge bowl. The spacing rib(s) may have a first portion that extends along the first surface, and a second portion that extends proximally along at least a portion of the tubular member. The plasmapheresis bowl and the first and second skirts may be configured such that the distance between the inner diameter of the proximal portion of the core wall and the outer diameter of the first and second skirts is maximized. The bowl may also include an extension tube connected to the tubular member at the distal end. The extension tube may extend toward the bottom of the plasmapheresis centrifuge bowl such that fluid entering the plasmapheresis centrifuge bowl is introduced nearer to the bottom of the plasmapheresis centrifuge bowl.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The foregoing features of the invention will be more readily understood by reference to the following detailed description, taken with reference to the accompanying drawings, in which:
- (2) FIG. 1 schematically shows a perspective view of a blood processing system in accordance with some embodiments of the present invention.
- (3) FIG. 2 schematically shows a top view of the blood processing system of FIG. 1, in accordance with some embodiments of the present invention;
- (4) FIG. 3 schematically shows a disposable set installed within a blood processing system, in accordance with some embodiments of the present invention.
- (5) FIG. 4 schematically shows a plasmapheresis centrifuge bowl for use with the system shown in FIGS. 1-3, in accordance with some embodiments of the present invention.
- (6) FIG. 5 schematically shows a cross section of the plasmapheresis centrifuge bowl shown in FIG. 4, in accordance with some embodiments of the present invention.
- (7) FIG. 6 schematically shows a cross section of the neck portion of the plasmapheresis centrifuge bowl shown in FIG. 4, in accordance with some embodiments of the present invention.
- (8) FIG. 7 schematically shows a feed tube for use in the plasmapheresis centrifuge bowl shown in FIG. 4, in accordance with various embodiments of the present invention.
- (9) FIGS. 8A-8E schematically show various views of a core for use in the plasmapheresis centrifuge bowl shown in FIG. 4, in accordance with various embodiments of the present invention.
- (10) FIGS. 9A-9B schematically show the plasmapheresis centrifuge bowl shown in FIG. 4 during blood processing, in accordance with various embodiments of the present invention.
- (11) FIG. 10 schematically shows a cross-sectional view of a plasmapheresis centrifuge bowl with an alternative core, in accordance with alternative embodiments of the present invention.
- (12) FIGS. 11A-11B schematically show various views of the alternative core shown in FIG. 10, in accordance with some embodiments of the present invention.
- (13) FIGS. 12A-12C schematically show various views of a further alternative core, in accordance

with some embodiments of the present invention.

DETAILED DESCRIPTION OF SPECIFIC EMBODIMENTS

(14) Illustrative embodiments of the present invention provide plasmapheresis bowls for the separation and collection of plasma. The bowl may have a core and a feed tube that increase the bowl efficiency and reduce foaming within the plasma. Details of the illustrative embodiments are discussed below.

(15) As shown in FIGS. 1 and 2, the blood processing system **100** includes a cabinet **110** that houses the main components of the system **100** (e.g., the non-disposable components). Within the cabinet **110**, the system **100** may include a first/blood pump **232** that draws whole blood from a subject, and a second/anticoagulant pump **234** that pumps anticoagulant through the system **100** and into the drawn whole blood. Additionally, the system **100** may include a number of valves that may be opened and/or closed to control the fluid flow through the system **100**. For example, the system **100** may include a donor valve **120** that may open and close to selectively prevent and allow fluid flow through a donor line **218** (e.g., an inlet line; FIG. 3), and a plasma valve **130** that selectively prevents and allows fluid flow through an outlet/plasma line **222** (FIG. 3). Some embodiments may also include a saline valve **135** that selectively prevents and allows saline to flow through a saline line **223**.

(16) To facilitate the connection and installation of a disposable set and to support the corresponding fluid containers, the system **100** may include an anticoagulant pole **150** on which the anticoagulant solution container **210** (FIG. 3) may be hung, and a saline pole **160** on which a saline solution container **217** (FIG. 3) may be hung (e.g., if the procedure being performed requires the use of saline). Additionally, in some applications, it may be necessary and/or desirable to filter the whole blood drawn from the subject for processing. To that end, the system **100** may include blood filter holder **170** in which the blood filter (located on the disposable set) may be placed.

(17) As discussed in greater detail below, apheresis systems **100** in accordance with embodiments of the present invention withdraw whole blood from a subject through a venous access device **206** (FIG. 3) using the blood pump **232**. As the system **100** withdraws the whole blood from the subject, the whole blood enters a blood component separation device **214**, such as a plasmapheresis centrifuge bowl **300** like that shown in FIG. 4, (a Latham type centrifuge or other type of separation chambers and devices may alternatively be used). The blood component separation device **214** separates the whole blood into its constituent components (e.g., red blood cells, white blood cell, plasma, and platelets). Accordingly, to facilitate operation of the separation device **214**, the system **100** may also include a well **180** in which the separation device **214** may be placed and in which the separation device **214** rotates (e.g., to generate the centrifugal forces required to separate the whole blood).

(18) To allow the user/technician to monitor the system operation and control/set the various parameters of the procedure, the system **100** may include a user interface **190** (e.g., a touch screen device) that displays the operation parameters, any alarm messages, and buttons which the user/technician may depress to control the various parameters. Additional components of the blood processing system **100** are discussed in greater detail below (e.g., in relation to the system operation).

(19) FIG. 3 is a schematic block diagram of the blood processing system **100** and a disposable collection set **200** (with an inlet disposable set **200A** and an outlet disposable set **200B**) that may be loaded onto/into the blood processing system **100**, in accordance with the present invention. The collection set **200** includes a venous access device **206** (e.g., a phlebotomy needle) for withdrawing blood from a donor's arm **208**, a container of anti-coagulant **210**, a centrifugation bowl **300** (e.g., a blood component separation device), a saline container **217**, and a final plasma collection bag **216**. The blood/inlet line **218** couples the venous access device **206** to an inlet port **330** of the bowl **300**, the plasma/outlet line **222** couples an outlet port **340** of the bowl **300** to the plasma collection bag **216**, and a saline line **223** connects the outlet port **340** of the bowl **300** to the saline container **217**.

An anticoagulant line **225** connects the anti-coagulant container **210** to the inlet line **218**. In addition to the components mentioned above and as shown in FIG. 3, the blood processing system **100** includes a controller **226**, a motor **228**, and a centrifuge chuck **230**. The controller **226** is operably coupled to the two pumps **232** and **234**, and to the motor **228**, which, in turn, drives the chuck **230**. The controller **226** may be operably coupled to and in communication with the user interface **190**.

(20) In operation, the disposable collection set **200** (e.g., the inlet disposable set **200A** and the outlet disposable set **200B**) may be loaded onto/into the blood processing system **100** prior to blood processing. In particular, the blood/inlet line **218** is routed through the blood/first pump **232** and the anticoagulant line **225** from the anti-coagulant container **210** is routed through the anticoagulant/second pump **234**. The centrifugation bowl **300** may then be securely loaded into the chuck **230**. Once the bowl **300** is secured in place, the technician may install the outlet disposable set **200B**. For example the technician may connect a bowl connector to the outlet **340** of the bowl **300**, install the plasma container **216** into the weight sensor **195**, run the saline line **223** through valve **135**, and run the plasma/outlet line **222** through valve **130** and the line sensor **185**. Once the disposable set **200** is installed and the anticoagulant and saline containers **210/217** are connected, the system **100** is ready to begin blood processing.

(21) As shown in FIG. 3, the system **100** may also include an optical sensor **213** that may be applied to a shoulder portion of the bowl **300**. The optical sensor monitors each layer of the blood components as they gradually and coaxially advance toward the core from the outer wall of the bowl **300**. The optical sensor **213** may be mounted in a position (e.g., within the well **180**) at which it can detect the buffy coat and/or the red blood cells reaching a particular radius, and, as discussed in greater detail below, the system **100** may alter the plasmapheresis in response to the detection.

(22) FIGS. 4 and 5 schematically show a perspective view and a cross-sectional view of a centrifuge bowl **300** (e.g., a plasmapheresis bowl) that may be used in conjunction with the system described above. The bowl **300** has an outer body **310** that defines the structure of the bowl **300** and an inner volume **320** into which the whole blood may be introduced for processing. The outer body **310**, in turn, includes a main wall **312**, a neck portion **316**, and shoulder portion **314** that connects the main wall **312** and the neck portion **316**. As discussed in greater detail below, the bowl **300** is rotatable about an axis in order to separate the whole blood into its various components (e.g., plasma, red blood cells, etc.).

(23) As discussed above, the bowl **300** may have an inlet **330** that allows whole blood to be introduced into the bowl **300**, and outlet port **340** that allows plasma (or other blood component) to be extracted from the bowl **300**. To allow the inlet **330** and outlet **340** to remain stationary during bowl rotations, and as best shown in FIGS. 5 and 6, the centrifuge bowl **300** may include a rotary seal **350** that connects the ports (e.g., the inlet **330** and outlet **340**) to the outer body **310** of the bowl **300**. The rotary seal **350** may include two rings (e.g., a ceramic ring **351A** and a carbon ring **351B**). One ring (e.g., the ceramic ring **351A**) is attached to the seal crown **356** which, in turn, is attached to the outer body **310**. The rotary seal **350** allows the bowl **300** (and a core **500** within the interior, discussed in greater detail below) to spin while the inlet **330** and outlet **340** remain stationary.

(24) In some embodiments, it may be beneficial to introduce the whole blood near the bottom of the bowl **300**. To that end, the bowl **300** may include a feed tube **400** that extends from the header assembly **355** of the bowl **300** into the interior **320** of the bowl **300**. As shown in FIG. 7, the feed tube **400** includes a tubular member **410** with a flow path **420** extending through it to allow the whole blood to flow through the feed tube **400**. One end of the tubular member **410** (e.g., the proximal end **412**) and the flow path **420** are fluidly connected to the inlet port **330**. At the distal end **414** of the tubular member **410**, the feed tube **400** has an extension tube **430** that extends from the tubular member **410**, through the core **500** (discussed in greater detail below) and toward the bottom of the bowl **300** (e.g., so that liquid flowing through the feed tube **400** is discharged at the

base **313** of the bowl body **310**).

(25) Nearer the distal end **414** of the tubular member **410**, the feed tube **400** has a skirt **440** that extends radially outward from the tubular member **410**. For example, the skirt **440** has a first portion **442** that extends generally perpendicularly out from the tubular member **410** such that one or more of the surfaces of the first portion **442** (e.g., the top surface **443**) is perpendicular to the longitudinal axis of the tubular member **410**. Extending from the first portion **442**, the skirt **440** may have an angled portion **444** that extends both radially outward from the first portion **442**, and downward/distally such that the top surface **445** of the angled portion **444** is angled downward (e.g., it is not perpendicular to the longitudinal axis of the tubular member **410**). It should be noted that the angled portion **444** may have a constant angle along the length or the angle may change gradually or in a step-wise fashion along the length of the angled portion **444**.

(26) The skirt **440** and a second skirt/disk **360** on the header assembly **355** of the bowl **300** may form an effluent channel **362** that is fluidly connected to the outlet **340** to allow blood components within the collection chamber **530** (discussed in greater detail below) to exit the bowl **300**. To create this effluent channel **362**, the skirt **440** may have a number (e.g., three) of spacing ribs **460** that maintain separation between the skirt **440** on the feed tube **400** and the skirt **360** on the header assembly **355** (e.g. to allow fluid to flow between the skirts **440/360**). The spacing ribs **460** may be located just on the top surface **443** of the first portion **442** or, as shown in FIG. 7, the spacing ribs **460** may extend along the top surface **443** and up the outer surface of the tubular member **410**. It should be noted that, although FIG. 7 shows three spacing ribs **460** that are equally spaced about the tubular member **410** and top surface **443**, other embodiments may have more or less than three spacing ribs **460**. Additionally or alternatively, the spacing ribs **460** may not be equally spaced about the tubular member **410** and top surface **443** (e.g., they may be irregularly spaced).

(27) As noted above, the header assembly **355** of the bowl **300** (including the inlet **330** and outlet **340**) does not rotate with the bowl **300**. Accordingly, because the feed tube **400** is connected to the header assembly **355**, it similarly does not rotate with the bowl **300**. Therefore, during collection of plasma via the effluent channel **362**, the stationary skirts **360/440** (e.g., creating the effluent channel) can impart a shear force on the spinning plasma component which, in turn, can cause foaming. However, by utilizing a smooth angled portion **444**/angled surface **445**, various embodiments of the present invention are able to reduce the shear force and, in turn, reduce foaming within the plasma. It should be noted that, because the spacing ribs **460** are located further down the effluent channel **362** and the rotation speed of the fluid/plasma has dropped by the time it reaches the ribs **460**, the spacing ribs **460** do not impart significant shear force/turbulence.

Additionally or alternatively, in some embodiments, the end **446** of the angled portion **444** (e.g., the outermost portion of the skirt **440**) may be rounded, chamfered or have other configurations that help reduce the shear force and/or help fluid enter the effluent channel **362**.

(28) As noted above, various embodiments of the present invention include a core **500** located within the interior of the bowl **300**. As shown in FIGS. 8A-8E, the core **500** may include a cylindrical body **510** that defines the overall structure of the core **500**. Within the interior of the cylindrical body **510**, the core **500** includes a ledge **520** that extends radially inward from the inner surface of the cylindrical body **510**. The portion of the cylindrical body **510** located above the ledge **520** (e.g., the proximal portion **512**) and the ledge **520** form a collection chamber **530** within the interior of the core **500** and in which the effluent channel **362** (formed by the skirts **360/440**) is located (e.g., so that the plasma may be extracted from the collection chamber **530**. To that end, the inner surface **514** of the proximal portion **512** (e.g., the wall formed by the proximal portion **512**) establishes the exterior boundary of the collection chamber **530**. The ledge **520** (e.g., the top surface **522** of the ledge **520**) establishes the lower boundary of the collection chamber **530** and creates a surface for incoming plasma to establish itself upon. The ledge **520** (e.g., the bottom surface **526** of the ledge **520**) also prevents fluid within the bowl **300** and below the core **500** from entering the collection chamber **530**.

(29) In addition to defining the collection chamber **530**, the proximal portion **512** of the core **500** also helps to isolate the collection chamber **530** from the separation chamber **325** within the bowl **300** (e.g., because fluid from the separation chamber **325** must flow up and over the wall of the proximal portion **512** to reach the collection chamber **530**). By isolating the collection chamber **530** from the separation chamber **325**, the proximal portion **512** prevents disrupting forces (e.g., shear forces, turbulent forces, etc.) created by the spinning fluid in the collection chamber **530** (e.g., plasma) contacting the stationary skirts **360/440** from entering the separation chamber **325**. By preventing these forces from reaching the separation chamber **325**, various embodiments of the present invention are able to maintain tighter packed cell layers (which would otherwise be disrupted by the turbulence that reaches the separation chamber **325**). This, in turn, increases collection efficiency and ensures that the signals from the optical sensor **213** are cleaner and more consistent (discussed in greater detail below).

(30) In addition improving the optical sensor **213** signals and allowing the cell layers to remain more tightly packed, the wall (e.g., the proximal portion **512**) also helps to reduce foaming within the plasma in several ways. For example, because the proximal portion **512** isolates the collection chamber **530** and the separation chamber **325**, the fluid (e.g., plasma) stays within the collection chamber **530** when the flow rates within the system **100** decrease or stop (e.g., due to whole blood pump **232** regulation by the system **100**). This, in turn, keeps the collection chamber **530** filled to a level where the skirts **360/440** (and the effluent channel **362**) remain wetted, and prevents air from mixing with the plasma. Additionally, the proximal portion **512** helps to form a stable fluid layer (comprised of the first volume of plasma entering the collection chamber **530**) on the inside wall of the proximal portion **512** in the collection chamber **530**. This stable fluid layer helps to stabilize the new/incoming fluid by residing below the entry point of the collection skirt **360/440**, and allowing the new plasma to “ride” on top and quickly exit through the collection skirt (e.g., the effluent channel **362**). This minimizes the mixing of the new/incoming fluid with air, reduces foam generation, and improves the line sensor **185** signal (which allows the system **100** to more accurately detect changes in cellular content exiting the bowl **300** and improves system efficiency).

(31) It should be noted that the location of the ledge **520** may impact the amount of foam generation in the bowl **300**. For example, if the ledge **520** is placed too high in the core **500**, the spinning core **500** will be located closer to the non-moving skirts **360/440**. This, in turn, would result in increased shear forces and foam generation. Additionally, if the ledge **520** is located too low, the plasma entering the collection chamber **530** would have a larger drop to the ledge **520** which can cause the plasma to “crash down” harder on the ledge **520** and increase turbulence and foam generation in the collection chamber **530**. Therefore, it is important that that ledge **520** is located far enough from the non-moving skirts **360/440** to minimize the shear forces and turbulence created between the ledge **520** and non-moving skirts **360/440**, but not so far that the plasma “crashes down” on the ledge **520**. A ledge **520** that is placed lower than the fluid fill level of a full non-spinning bowl **300** may allow the surfaces of the collection chamber **530** to become submerged in red cells during the beginning of return, and after the bowl **300** has fully drained there could still be residual red cells left on the interior surfaces of the core (e.g., on the surface of the proximal wall **512** and on the top surface **522** of the ledge **520**) that could be picked up by the plasma of the subsequent cycle when it enters the collection chamber **530**.

(32) The proximal portion **512** (e.g., the wall) also helps to prevent red blood cells within the separation chamber **325** from entering the collection chamber **530** when the draw cycle is complete (e.g. when the whole blood pump **232** stops and the centrifuge (e.g., the motor **228** and chuck **230** slow down). As the bowl **300** slows down, there is a decreasing centrifugal force on the blood components of the bowl **300**, and as such the separation of blood into cellular layers will be lost. Thus, red cells and other cells that were previously being packed to the outside of the bowl body **310** may mix in with the plasma still within the separation chamber **325**. The proximal portion **512** of the core wall may contain this mixing of cellular components with the plasma to the separation

chamber **325**, and may reduce the entrance of cells from the separation chamber **325** into the collection chamber **530**. This allows the plasma within the collection chamber **530** to remain separate from the red blood cells (and other components) even as the bowl **300** slows down. When the bowl **300** is fully drained, there are potentially less red blood cells and other components left as a residue on the interior portion of the core (e.g., on the proximal wall **512** and top surface **522** of the ledge **520**), resulting in less cells being picked up by the entering plasma of the subsequent cycle. Therefore, the amount of red blood cells inadvertently collected during plasmapheresis (e.g., between cycles) is greatly reduced.

(33) The portion of the cylindrical body **510** located below the ledge **520** (e.g., the portion **516**) provides a reflective surface for the optical sensor **213**. In other words, during processing, the optical sensor **213** on the bowl will shine a light into the bowl and the amount of transmission/reflection back to the sensor **213** provides an indication of material layer locations within the bowl **300**. The portion **516** of the core **500** below the ledge **520** provides the surface which the light reflects off of and back toward the sensor **213**.

(34) In some embodiments, the core **500** may have a constant diameter along its length (e.g., with the exception of the ribs **540**, the core **500** may have a smooth/vertical outer surface **513**) to ensure that there are no overhangs in which the plasma may be trapped (e.g., because fluid will fill to the diameter of the skirt and is unable flow against the centrifugal force created by the spinning bowl **300**). Additionally, if the diameter of any overhang is smaller than the skirts **360/440**, the overhang may also cause air to be trapped. This causes a disruption of the optics signal, if the trapped air is in the path of reflectance.

(35) As best shown in FIGS. **8C-8E**, the ledge **520** may have an opening **524** (e.g., a hole) extending through the center of it. This opening **524** allows the extension tube **430** of the feed tube **400** to pass through the core **500** and the ledge **520** and provides an opening through which fluid remaining within the collection chamber **530** at the end of processing and plasma collection may exit the collection chamber **530** and drain to the bottom of the bowl **300**, for example, so it can be returned to the patient or sent to a separate collection container. To help with the drainage and minimize fluid hold up in the collection chamber **530**, the ledge **520** may extend distally (e.g., downward) toward the bottom of the core **500** such that the top surface **522** slopes toward the opening **524**.

(36) To position and secure the core **500** within the interior of the bowl **300**, the core **500** may have a number of ribs **540** near the proximal end (e.g., the top) of the core **500**. For example, the bottom surface **542** of the ribs **540** may interface with a mating ledge **317** (FIGS. **5** and **6**) on the bowl body **310**, and the outer surface **544** of the ribs **540** may interface with the interior face **318** of the neck portion **316** of the bowl **300**/bowl body **310**. In some embodiments, the interface between the outer surface **544** of the ribs **540** and the interior face **318** of the neck portion **316** may create an interference fit between the bowl **300**/bowl body **310** and core **500**. To further hold the core **500** in place within the bowl **300** and maintain the vertical location of the core **500** within the bowl **300**, the top surface **546** of the ribs **540** may also interface with a seal crown **356** on the header assembly **355**.

(37) The ribs **540** may extend proximally (upwards) past the proximal end (e.g., top) of the cylindrical body **510** of the core and may be spaced around the diameter of the cylindrical body **510**. In this matter, the ribs **540** create flow channels **550** between them that allow fluid to flow over the proximal portion **512**, between the ribs **540** and into the collection chamber **530**. It should be noted that, although FIGS. **8A-8E** shows eight ribs **540** that are equally spaced about the core **500**, other embodiments may have more or less than eight ribs **540**. Additionally or alternatively, the ribs **540** may not be equally spaced about the core **500** (e.g., they may be irregularly spaced).

(38) It should also be noted that the sizing and positioning of the core **500** within the bowl **300** may impact the amount of shear force created between the core **500** (which spins during processing) and the skirts **360/440** (which are stationary). Therefore, to minimize the amount of shear forces created

(and, as a result, the amount of foam within the plasma), various embodiments of the present invention may maximize the distance between the inner surface **514** of the core **500** and the skirts **360/440**. This may be accomplished in several ways. For example, in some embodiments, the thickness of the wall of the bowl **300** may be reduced in the neck portion **316** (e.g., by increasing the inner diameter of the neck portion **316**). This allows the core **500** to be wider and, therefore, the distance between the core **500** and skirts **360/440** to be increased. Additionally, by increasing the distance in this manner, other aspects of the bowl **300** (e.g., the overall exterior shape and dimensions of the bowl **300**, the location of the bowl body weld, the location of the optics signal transmission, etc.) may remain the same, and the bowl **300** may still be used with existing plasmapheresis systems.

(39) During use and after lines **222/223** are connected and the bowl **300** is installed into the system **100**, the user/technician may insert the venous access device **206** into the donor's arm **208** and the controller **226** may activate the two pumps **232, 234** and the motor **228**. Operation of the two pumps **232, 234** causes whole blood to be drawn from the donor, anticoagulant from container **210** to be introduced into the drawn whole blood, and the now anticoagulated whole blood to be delivered to the inlet port **330** of the bowl **300**.

(40) It should be noted that the anticoagulant line **225** may also include a bacteria filter (not shown) that prevents any bacteria in the anticoagulant source **210**, the anticoagulant, or the anticoagulant line **225** from entering the system **100** and/or the subject. Additionally, the anticoagulant line **225** may include an air detector **140** that detects the presence of air within the anticoagulant. The presence of air bubbles within any of the system **100** lines can be problematic for the operation the system **100** and may also be harmful to the subject if the air bubbles enter the blood stream. Therefore, the air detector may be connected to an interlock that stops the flow within the anticoagulant line **225** in the event that an air bubble is detected (e.g., by stopping the anticoagulant pump **234**), thereby preventing the air bubbles from entering the subject.

(41) As the anti-coagulated whole blood is withdrawn from the subject and introduced into the plasmapheresis bowl **300**. The whole blood will flow through the feed tube **400** and extension tube **430** and into the bowl **300** near the bottom **313** of the bowl **300**. The centrifugal forces caused by the bowl rotation will cause the whole blood to move toward the outer wall **312** of the bowl **300**, and the blood component separation device **214** (e.g., the bowl **300**) will separate the whole blood into several blood components. For example, the bowl **300** may separate the whole blood into a first, second, third, and, perhaps, fourth blood component. More specifically, the bowl **300** (and the centrifugal forces created by rotation of the bowl **300**) can separate the whole blood into plasma, platelets, red blood cells ("RBC"), and, perhaps, white blood cells ("WBC"). The higher density component, i.e., RBC **580**, is forced to the outer wall of the bowl **300** while the lower density plasma **590** lies nearer the core **500**. A buffy coat **585** is formed between the plasma and the RBC. The buffy coat **585** is made up of an inner layer of platelets, a transitional layer of platelets and WBC and an outer layer of WBC.

(42) As shown in FIG. 3 and as briefly discussed above, the system **100** may also include an optical sensor **213** that may be applied to a shoulder portion **314** of the bowl **300**. The optical sensor **213** monitors each layer of the blood components as they gradually and coaxially advance toward the core from the outer wall of the bowl **300**. The optical sensor **213** may be mounted in a position (e.g., within the well **180**) at which it can detect the buffy coat and/or the red blood cells reaching a particular radius, and the steps of drawing the whole blood from the subject/donor and introducing the whole blood into the bowl **300** may be altered and/or terminated in response to the detection.

(43) Once the bowl **300** has separated the blood into the various components (e.g., red blood cells **580** and plasma **590**, FIGS. 9A and 9B), one or more of the components can be removed from the bowl **300**. For instance, as additional anticoagulated whole blood enters the bowl **300**, the plasma **590** will be forced further inward toward the core **500** until it flows over the proximal portion **512** of the core **500** and into the collection chamber **530**. When the collection chamber **530** fills with

plasma such that the plasma within the collection chamber **530** makes sufficient contact with the skirts **360/440** and effluent channel **362** (FIGS. **9A** and **9B**), the plasma will begin exiting the bowl **300** via the effluent channel **362** and the outlet **340**. Once out of the bowl **300**, the plasma will flow through line **222** and into the plasma collection container **216**. Some embodiments of the system **100** may include a weight sensor **195** (FIG. **1**) that measures the amount of plasma collected. The plasma collection process may continue until a target or pre-determined volume of plasma is collected within the plasma collection container **216**.

(44) As noted above, in some embodiments, the system **100** may also include a line sensor **185** that can determine the type of fluid (e.g., plasma, platelets, red blood cells etc.) exiting the bowl **300**. In particular, the line sensor **185** consists of an LED which emits light through the blood components leaving the bowl **300** and a photo detector which receives the light after it passes through the components. The amount of light received by the photo detector is correlated to the density of the fluid passing through the line. For example, if plasma is exiting the bowl **300**, the line sensor **185** will be able to detect when the plasma exiting the bowl **300** becomes cloudy with platelets (e.g., the fluid exiting the bowl **300** is changing from plasma to platelets). The system **100** may then use this information to either stop the removal of blood components from the bowl **300**, stop drawing whole blood from the subject, or redirect the flow by, for example, closing one valve and opening another.

(45) Once the system **100** has collected the target volume of plasma within the plasma collection container **216** or the bowl **300** is full of red blood cells, the system **100** can return the remaining components (e.g., the components remaining within the bowl **300**) to the subject. For example, when all the plasma has been removed and the bowl **300** is full of RBCs (and any other blood component not collected), the controller **226** stops the draw of whole blood from the subject and slows and/or stops the bowl **300**. As the bowl **300** slows and/or stops, and fluid remaining within the collection chamber **530** will drain out of the collection chamber **530** via the opening **524** in the ledge **520**. The system **100**/controller **226** may then reverse the direction of the blood/first pump **232** to draw the RBCs (and other components) from the bowl **300** and send them back to the subject. Alternatively, if the system **100** is so equipped, the system **100** may return the components to the subject via a dedicated return line.

(46) In addition to the non-collected blood components (e.g., the components remaining in the bowl **300**), the system **100** may also return saline to the patient/subject. The saline may be used as a compensation fluid to make up for the volume of the blood component (e.g., plasma) that was removed and collected, and is not being returned to the patient. To that end, during the return step, the saline valve **135** may be opened to allow saline from the saline container **217** to flow through the saline line **223** and into the bowl **300** (via outlet **340**), where it can be returned to the patient/donor with or after the remaining blood components. If additional plasma collection cycles are to be performed (e.g., if the volume of plasma already collected does not equal the target/pre-determined volume), the system **100** may once again start the blood/first pump **232** to withdraw whole blood from the subject and the system **100** may repeat the process above until the target volume of plasma is collected.

(47) FIG. **10** schematically shows a plasmapheresis bowl **300** with an alternative embodiment of the core (e.g., a “long” core **600**). FIGS. **11A** and **11B** schematically show a perspective view and a cross-sectional view of the alternative core **600**. As can be seen in FIGS. **11A** and **11B**, the “long” core **600**, like the core **500** shown in FIGS. **8A-8E**, has a cylindrical body **610** that defines the overall structure of the core **600**, and a ledge **520** that extends radially inward from the inner surface of the cylindrical body **610**. Also like the core **500** shown in FIGS. **8A-8E**, the proximal portion **612** of the body **610** (e.g., the inner surface **614** of the proximal portion **612**) located above the ledge **520** and the ledge **520** form the collection chamber **530**. The core **600** also has a number of ribs **640** that position and secure the core **600** within the bowl **300** in a similar manner as that described above. For example, the bottom surface **642** of the ribs **640** may interface with the

mating ledge **317** (FIG. **10**) on the bowl body **310**, the outer surface **644** of the ribs **640** may interface with the interior face **318** of the neck portion **316** of the bowl **300**, and the top surface **646** of the ribs **640** may interface with the seal crown **356** on the header assembly **355** (e.g., to maintain the vertical location within the bowl **300**). The spaces between the ribs **640** also create flow channels **650** that allow fluid to flow between the ribs **640** and into the collection chamber **530**.

(48) Unlike the core **500** shown in FIGS. **8A-8E**, the portion of the long core **600** embodiment below the ledge **520** is significantly longer and extends further downward into the bowl **300**. For example, directly under the ledge **520**, the core **600** may include an intermediate portion **615** that performs a function similar to that described above for the portion **516** of the core **500** shown in FIGS. **8A-8E**. In particular, the intermediate portion **615** may provide reflective surface for the optical sensor **213**. Directly below the intermediate portion **615**, the core **600** may have an extended distal portion **616** that extends further downward into the bowl **300**.

(49) In some instances, this long core **600** can provide some benefits over the shorter version of the core. In particular, during processing, fluid may become trapped in the space under the collection chamber **530** and ledge **520** and inside of the wall of portion **516**. However, by extending further into the bowl **300** and taking up more volume within the interior of the bowl **300**, the extended distal portion **616** of the long core **600** helps to prevent some of the fluid/plasma from becoming trapped, which allows more plasma to exit the bowl and increases the collection efficiency. Although the extended distal portion **616** is shown as a relatively thin wall, other embodiments may have a thick walled extended distal portion **616** (under the collection chamber **530**) to take up more volume under the collection chamber **530**, which may prevent even more fluid/plasma from being trapped. Alternatively, the core **600** may have a second ledge (not shown) extending radially inward from the bottom of the extended/distal portion **616** to prevent fluid from traveling inside of the distal portion of the core **600**.

(50) The extended distal portion **616** of the long core **600** also helps to stabilize the plasma layer within the separation chamber **325**. For example, as noted above, as the whole blood separates into its various components, the plasma is located nearest to the center of the bowl **300** (e.g., nearest the core **500/600**). The extended distal portion **616** of the long core **600** provides a solid contact surface for the plasma (which is spinning at the same rate as the core **600**), as opposed to the air cylinder (e.g., the area radially inward from the plasma layer that is taken up by air rather than a blood component). By providing the plasma with a solid contact surface, the extended distal portion **616** of the long core **600** helps to diffuse any turbulent or shearing forces that propagate from the collection chamber **530** and into the separation chamber **325**. Furthermore, by stabilizing the component layers and diffusing any turbulent or shearing forces, the long core **600** may increase collection efficiency and the quality of the collected plasma (e.g., less cellular components mix with the plasma).

(51) Although the cores **500/600** described above have a number of ribs **540/640** that position the cores **500/600** within the bowl **300**, other embodiments of the plasmapheresis bowl **300** may utilize cores with different structures. For example, as shown in FIGS. **12A-12C**, some embodiments of the core (e.g., core **700**) can include a ring **740** that is concentric with the tubular body **710** of the core **700**. In a similar manner to the various surfaces of the ribs **540/640**, the bottom surface **742** of the ring **740** may interface with the mating ledge **317** on the bowl body **310**, the outer surface **744** of the ring **740** may interface with the interior face **318** of the neck portion **316** of the bowl **300**, and the top surface **746** of the ring **740** may interface with the seal crown **356** on the header assembly **355**.

(52) To allow fluid/plasma to pass between the ring **740** and the tubular body **710** (e.g., so that it may flow up and over the proximal portion **712** and into the collection chamber **530**), the core **700** may include spacers **755** that are spaced about the tubular body **712**. The spacers **755** space the ring **740** from the tubular body **710** and provide flow paths **750** between the ring **740** and tubular body **710**. It should be noted that, although the ring **740** is shown on a “long” version of the core, the

ring 740 can also be utilized on the shorter core (e.g., like those shown in FIGS. 8A-8E) or a core having a length between the long and short core versions.

(53) It is important to note that, although the various embodiments discussed above are in relation to a blood processing system that collects plasma, the features discussed herein may be applied to any type of blood processing system. For example, the features described herein may be implemented on blood processing systems that collect and/or process red blood cells, platelets and/or white blood cells.

(54) The embodiments of the invention described above are intended to be merely exemplary; numerous variations and modifications will be apparent to those skilled in the art. All such variations and modifications are intended to be within the scope of the present invention as defined in any appended claims.

Claims

1. A feed tube for a plasmapheresis centrifuge bowl comprising: a tubular member extending between a proximal end and a distal end, the tubular member having a flow path extending there through, the flow path configured to fluidly connect an inlet port on the plasmapheresis centrifuge bowl and an interior of the plasmapheresis centrifuge bowl; and a first skirt member extending radially outward from the tubular member, the first skirt member having a first surface generally perpendicular to the tubular member and an angled surface extending radially outward and distally from the first surface, the entirety of the angled surface being smooth such that the angled surface is free from protrusions.
2. A feed tube according to claim 1, further comprising: at least one spacing rib located on the first surface and configured to space the first skirt member from a second skirt member, thereby creating a fluid channel extending between the first skirt member and the second skirt member.
3. A feed tube according to claim 2, wherein the at least one rib has a first portion that extends along the first surface, and a second portion that extends proximally along at least a portion of the tubular member.
4. A feed tube according to claim 2, wherein the at least one rib includes three ribs.
5. A feed tube according to claim 4, wherein the three ribs are equally spaced about the first skirt member.
6. A feed tube according to claim 2, wherein the fluid channel is configured to fluidly connect a collection chamber within the plasmapheresis centrifuge bowl with an outlet port on the plasmapheresis centrifuge bowl.
7. A feed tube according to claim 1, further comprising an extension tube connected to the tubular member at the distal end and configured to extend toward a bottom of the plasmapheresis centrifuge bowl, such that fluid entering the plasmapheresis centrifuge bowl via the feed tube is introduced nearer to the bottom of the plasmapheresis centrifuge bowl.
8. A plasmapheresis centrifuge bowl comprising: an outer body rotatable about a longitudinal axis of the plasmapheresis centrifuge bowl, the outer body having a main body defining an interior cavity, a neck portion extending proximal to the main body, and a shoulder connecting the main body and the neck portion; a separation region located within the interior cavity, rotation of the plasmapheresis centrifuge bowl separating whole blood within the separation region into a first blood component and a second blood component; an inlet port for introducing whole blood into the plasmapheresis centrifuge bowl; a feed tube having: a tubular member extending between a proximal end and a distal end, the tubular member having a flow path extending there through, the flow path configured to fluidly connect the inlet port and the interior of the plasmapheresis centrifuge bowl, and a first skirt member extending radially outward from the tubular member, the first skirt member having a first surface generally perpendicular to the tubular member and an angled surface extending radially outward and distally from the first surface, the entirety of the

angled surface being smooth such that the angled surface is free from protrusions; an outlet port for extracting a first blood component out of the plasmapheresis centrifuge bowl; and a rotary seal fluidly coupling the inlet port and outlet port to the outer body.

9. A plasmapheresis centrifuge bowl according to claim 8, wherein the feed tube further includes: at least one spacing rib located on the first surface and configured to space the first skirt member from a second skirt member, thereby creating a fluid channel extending between the first skirt member and the second skirt member.

10. A plasmapheresis centrifuge bowl according to claim 9, wherein the at least one spacing rib has a first portion that extends along the first surface, and a second portion that extends proximally along at least a portion of the tubular member.

11. A plasmapheresis centrifuge bowl according to claim 9, wherein the at least one spacing rib includes three ribs.

12. A plasmapheresis centrifuge bowl according to claim 11, wherein the three spacing ribs are equally spaced about the first skirt member.

13. A plasmapheresis centrifuge bowl according to claim 9, wherein the second skirt member is located on a header assembly of the plasmapheresis centrifuge bowl.

14. A plasmapheresis centrifuge bowl according to claim 13, further comprising a core located within and rotatable with the outer body, the core having a cylindrical body defining the core and an interior of the core, and a ledge located within the interior of the core between a proximal end and a distal end of the cylindrical body, the ledge extending radially inward from an inner diameter of the core and having a top surface that defines, at least partially, a collection chamber within the plasmapheresis centrifuge bowl.

15. A plasmapheresis centrifuge bowl according to claim 14, wherein the core and the first and second skirts are configured such that the distance between the inner diameter of a proximal portion of the core and the outer diameter of the first and second skirts is maximized.

16. A plasmapheresis centrifuge bowl according to claim 14, wherein the ledge includes an opening extending therethrough, the feed tube extending through the opening.

17. A plasmapheresis centrifuge bowl according to claim 14, wherein the fluid channel fluidly connects the collection chamber and the outlet port.

18. A plasmapheresis centrifuge bowl according to claim 8, wherein the feed tube further includes an extension tube connected to the tubular member at the distal end and extending toward a bottom of the plasmapheresis centrifuge bowl, such that fluid entering the plasmapheresis centrifuge bowl is introduced nearer to the bottom of the plasmapheresis centrifuge bowl.
